

Boolean Algebra & Simplification

Boolean Algebra:

→ Boolean algebra is a mathematics of digital logic.

→ It is used to perform the logical operations in digital computers.

→ In digital computers True represent by '1' (high volt) and False represent by 0 (low volt).

Basic Theorem of Boolean Algebra:

T₁: Properties of 0

$$0 + A = A$$

$$0 \cdot A = 0$$

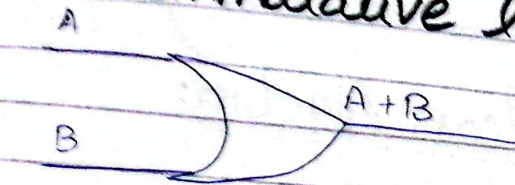
T2: Properties of 1

$$1 + A = 1$$

$$1 \cdot A = A$$

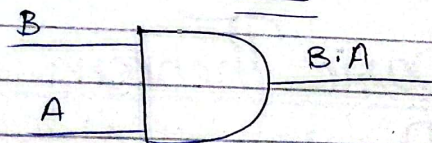
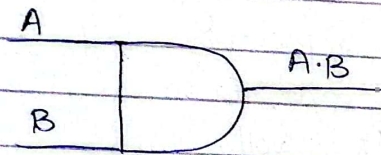
T3: Commutative law

OR:-



$$A + B = B + A$$

AND:-

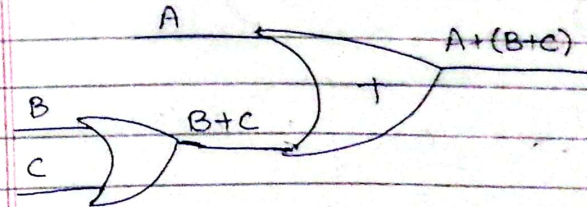


$$A \cdot B = B \cdot A$$

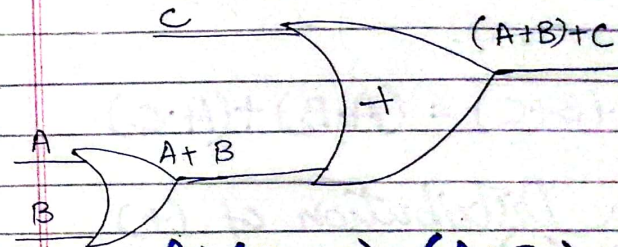
Redundancy or Consensus Theorem:

$$Y = AB + \bar{B}C + AC = AB + \bar{B}C$$

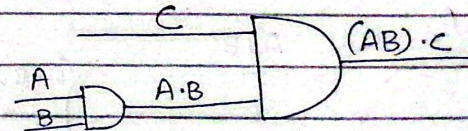
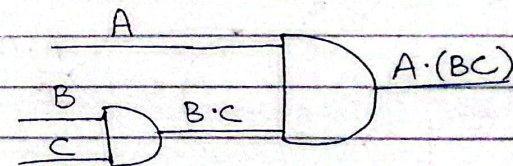
Associative law:



$$A + (B + C)$$



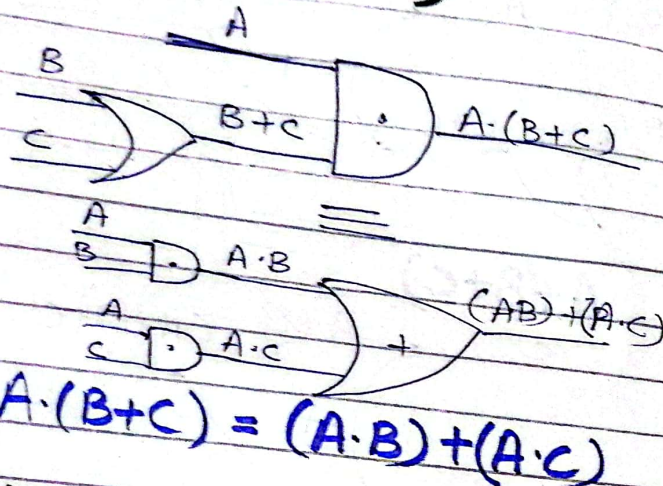
$$A + (B + C) = (A + B) + C$$



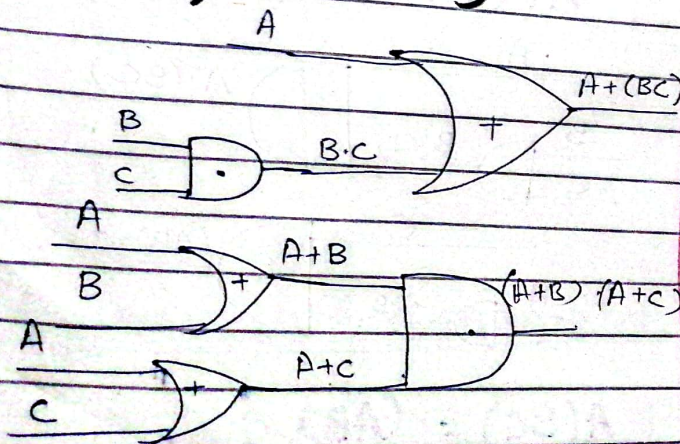
$$A(B \cdot C) = (A \cdot B) \cdot C$$

Distributive law:

(a) Distribution of $(\cdot)$ over $(+)$



(b) Distribution of $(+)$ over $(\cdot)$



T6 - Idempotence Law
 $A + A = A$
 $A \cdot A = A$

T7 - Absorption Law
 $A + A \cdot B = A$
 $A \cdot (A + B) = A$

T8 - Complementary Law
 $X + X' = 1$
 $X \cdot X' = 0$

T9 - Involution

$$X'' = X$$

T10 - De Morgan's Theorem
Boolean Addition:

Boolean addition is equivalent to OR operation.

$$0 + 0 = 0$$

$$0 + 1 = 1$$

$$1 + 0 = 1$$

$$1 + 1 = 1$$

Boolean Multiplication:

$$0 \cdot 0 = 0$$

$$0 \cdot 1 = 0$$

$$1 \cdot 0 = 0$$

$$1 \cdot 1 = 1$$

$$A \cdot A = A$$

$$A \cdot \bar{A} = 0$$

Determine the values of A, B, C and D that make sum term $A + \bar{B} + C + \bar{D}$ equal to 0.

for the sum term to be 0, each of the literals in the sum must be 0.

Therefore, $A = 0, B = 1, C = 0, \bar{B} = 0, D = 1, \bar{D} = 0$

$$= A + \bar{B} + C + \bar{D}$$

$$= 0 + \bar{1} + 0 + \bar{1}$$

$$= 0 + 0 + 0 + 0$$

$$= 0$$

Determine the values of A, B, C and D that make the product term $A\bar{B}C\bar{D}$ equal to 1.

for the product term to be 1, each of the literal term must be 1.

$A = 1, B = 0$ so that $\bar{B} = 1, C = 1, D = 0$, so that $\bar{D} = 1$

$$= A\bar{B}C\bar{D}$$

$$= 1 \times 1 \cdot 1 \cdot 1$$

$$= 1$$

if $A = 0$, what does $\bar{A}$ equal?

$$A = 0 = 1$$

Determine the values of A, B and C that make the sum term $\bar{A} + B + C$ equal to 0?

$$A=1, B=1, C=0, \bar{A}=0, \bar{B}=0$$

$$\bar{A} + \bar{B} + C$$

$$0 + 0 + 0$$

Determine the values of A, B, C to make the product term 1.

$$A=1, B=0, C=1$$

$$\bar{B}=1$$

$$A \cdot \bar{B} \cdot C$$

$$= 1 \times 1 \cdot 1$$

$$= 1$$

Prove that $A + \bar{A}B = A + B$.

$$A + \bar{A}B = A + AB + \bar{A}B \quad \text{Rule 10} = A + B$$

$$= A + B(A + \bar{A})$$

$$= A + B(1)$$

$$= A + B$$

Q# Define duality property?
Duality property states that every algebraic expression derivable from the postulates of boolean algebra remains valid if the operator and identity elements are interchanged.

Rule 12: $(A+B)(A+C) = A+BC$

$$(A+B)(A+C) = A \cdot A + AC + BA + BC$$

$$= A + AC + AB + BC$$

$$= A(1+C) + AB + BC$$

$$= A(1) + AB + BC$$

$$= A + AB + BC$$

$$= A(1+B) + BC$$

$$= A \cdot 1 + BC$$

$$= A + BC$$

Rule 10:

$$A + AB = A$$

$$= A + AB$$

$$= A(1+B) \quad \because 1+B=1$$

$$= A(1) \quad \because A \cdot 1 = 1$$

$$= A$$

Apply associative law of addition to the expression $A + (B + C + D)$

$$= A + (B + C + D) = (A + B + C) + D$$

Apply distributive law $A(B+C+D)$

$$A(B+C+D) = AB + AC + AD$$

De Morgan's Theorem

De Morgan's first theorem is stated as follows:

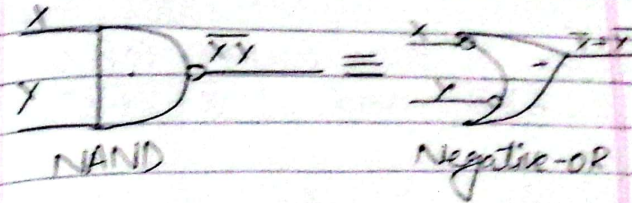
→ The complement of a product of variables is equal to the sum of the complement of the variables.

$$\overline{XY} = \bar{X} + \bar{Y}$$

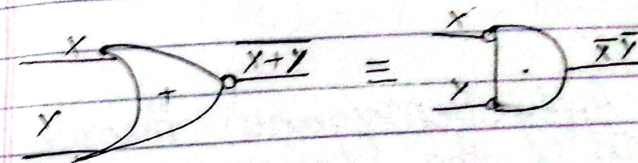
De Morgan's second theorem is stated as follows:

→ The complement of a sum of variables is equal to the product of the complement of the variables.

$$\overline{X+Y} = \bar{X} \cdot \bar{Y}$$



X	Y	$\bar{X}$	$\bar{Y}$	$\overline{XY}$	$\bar{X} + \bar{Y}$
0	0	1	1	1	1
0	1	1	0	1	1
1	0	0	1	1	1
1	1	0	0	0	0



X	Y	$\bar{X}$	$\bar{Y}$	$\overline{X+Y}$	$\bar{X} \cdot \bar{Y}$
0	0	1	1	1	1
0	1	1	0	0	0
1	0	0	1	0	0
1	1	0	0	0	0

Apply DeMorgan's theorem to the expressions $\overline{XYZ}$ and $\overline{X+Y+Z}$

$$\overline{XYZ} = \overline{X+Y+Z}$$

$$\overline{XYZ} = \overline{X+Y+Z}$$

$$\overline{X+Y+Z} = \overline{X} \cdot \overline{Y} \cdot \overline{Z}$$

Apply DeMorgan's theorem to the expression $\overline{X+Y+Z}$

$$= \overline{X} \cdot \overline{Y} \cdot \overline{Z}$$

$$= XYZ$$

Apply DeMorgan's theorem to the expressions $\overline{WXYZ}$ and $\overline{W+X+Y+Z}$

$$\overline{WXYZ} = \overline{W+X+Y+Z}$$

$$\overline{W+X+Y+Z} = \overline{W} \cdot \overline{X} \cdot \overline{Y} \cdot \overline{Z}$$

Apply DeMorgan's theorem to the expression $\overline{W} \cdot \overline{X} \cdot \overline{Y} \cdot \overline{Z}$

$$= \overline{\overline{W} + \overline{X} + \overline{Y} + \overline{Z}}$$

$$= \overline{W+X+Y+Z}$$

$$= W+X+Y+Z$$

Applying DeMorgan's Theorems:-

$$\overline{A+B\overline{C}+D(E+\overline{F})}$$

$$\text{Let } A+B\overline{C} = X \text{ and } D(E+\overline{F}) = Y$$

$$= \overline{A+B\overline{C}+D(E+\overline{F})}$$

$$= \overline{A+B\overline{C}} \cdot \overline{D(E+\overline{F})}$$

$$= (\overline{A+B\overline{C}}) \cdot \overline{D(E+\overline{F})}$$

$$= (\overline{A+B\overline{C}}) \cdot (\overline{D} + \overline{E+\overline{F}})$$

$$= (\overline{A+B\overline{C}}) \cdot (\overline{D} + \overline{E} + \overline{\overline{F}})$$

Example 4-5

Applying DeMorgan's theorem to each of the following expressions.

$$(a) \overline{(A+B+C)D}$$

$$\text{Let } A+B+C = X \text{ and } D = Y$$

$$= \overline{A+B+C} + \overline{D}$$

$$= \overline{A} \cdot \overline{B} \cdot \overline{C} + \overline{D}$$

$$\begin{aligned}
 & \overline{ABC + DEF} \\
 & \text{Let } ABC = X \text{ and } DEF = Y \\
 & = \overline{ABC} \cdot \overline{DEF} \\
 & = (\overline{A + B + C}) (\overline{D + E + F})
 \end{aligned}$$

$$\begin{aligned}
 & \overline{AB} + \overline{CD} + EF \\
 & = (\overline{AB}) (\overline{CD}) (\overline{EF}) \\
 & = (\overline{A + B}) (\overline{C + D}) (\overline{E + F})
 \end{aligned}$$

Related problem:

$$\begin{aligned}
 & \overline{ABC} + D + E \\
 & = (\overline{ABC}) (\overline{D}) (\overline{E}) \\
 & = (\overline{A + B + C}) (\overline{D}) (\overline{E}) \\
 & = (\overline{A + B + C}) (\overline{D}) (\overline{E})
 \end{aligned}$$

Example 4-6
Applying DeMorgan's theorem:-

$$\begin{aligned}
 (a) \quad & \overline{(A + B) + C} \\
 & = (\overline{A + B}) (\overline{C}) \\
 & = (\overline{A + B}) \overline{C}
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad & \overline{(A + B) + CD} \\
 & = (\overline{A + B}) (\overline{CD}) \\
 & = (\overline{A + B}) (\overline{C + D}) \\
 & = (\overline{A + B}) (\overline{C + D})
 \end{aligned}$$

$$\begin{aligned}
 & (A + B) \overline{C} \overline{D} + E + \overline{F} \\
 & = ((A + B) \overline{C} \overline{D}) (E + \overline{F}) \\
 & = (\overline{A + B + C + D}) (\overline{E} \cdot F)
 \end{aligned}$$

Related problem:

$$\begin{aligned}
 & \overline{AB} (C + \overline{D}) + E \\
 & = (\overline{AB} (C + \overline{D})) (\overline{E}) \\
 & = (\overline{A + B}) (\overline{C + D}) (\overline{E}) \\
 & = (\overline{A + B + C + D}) (\overline{E})
 \end{aligned}$$

Example 4-7

Boolean expression of an exclusive OR gate is $\overline{A}B + A\overline{B}$.

Apply DeMorgan's theorem

$$\begin{aligned}
 & = \overline{A\overline{B} + \overline{A}B} \\
 & = (\overline{A\overline{B}}) (\overline{\overline{A}B}) \\
 & = (\overline{A + B}) (\overline{\overline{A} + \overline{B}}) \\
 & = (\overline{A + B}) (A + B) \\
 & = A\overline{A} + \overline{A}\overline{B} + AB + B\overline{B} \\
 & = 0 + \overline{A}\overline{B} + AB + 0 \\
 & = \overline{A}\overline{B} + AB
 \end{aligned}$$

Section 4-3

1) Apply De Morgan's theorem to the following expressions.

$$(a) \overline{ABC} + \overline{D+E}$$

$$= (\overline{ABC})(\overline{D+E})$$

$$= \overline{A} + \overline{B} + \overline{C} + \overline{D} + \overline{E}$$

$$(b) \overline{(A+B)C}$$

$$= \overline{A} \overline{B} + \overline{C}$$

$$(c) \overline{A+B+C} + \overline{DE}$$

$$= \overline{A} \overline{B} \overline{C} + \overline{D} + \overline{E}$$

Example 4-9

Using Boolean algebra techniques, simplify this expressions.

$$AB + A(B+C) + B(B+C)$$

$$= AB + AB + AC + B \cdot B + BC$$

$$= AB + AB + AC + B + BC$$

$$= AB + AB + AC + B(1+C)$$

$$= AB + AB + AC + B(1)$$

$$= AB + AC + AB + B$$

$$= AB + AC + B(1+A)$$

$$= AB + AC + B(1)$$

$$\begin{aligned}
 & AB + AC + B \\
 &= B(1+A) + AC \\
 &= B(1) + AC \\
 &= B + AC
 \end{aligned}$$

Simplify the Boolean expression.

$$\begin{aligned}
 & A\bar{B} + A(\overline{B+C}) + B(\overline{B+C}) \\
 &= A\bar{B} + A(\bar{B}\bar{C}) + B(\bar{B}\bar{C}) \\
 &= A\bar{B} + A\bar{B}\bar{C} + B\bar{B}\bar{C} \\
 &= A\bar{B} + A\bar{B}\bar{C} + \bar{C}(0) \\
 &= A\bar{B} + A\bar{B}\bar{C} \\
 &= \bar{B}(A + A\bar{C}) \Rightarrow A\bar{B}(1 + \bar{C}) = A\bar{B}(1) \\
 &= A\bar{B}
 \end{aligned}$$

Example 4-10

Simplify the following Boolean expression

$$[A\bar{B}(C+BD) + \bar{A}\bar{B}]C$$

Apply distributive theorem:-

$$\begin{aligned}
 &= [A\bar{B}C + AB\bar{B}D + \bar{A}\bar{B}]C \\
 &= A\bar{B}C + 0 + \bar{A}\bar{B}]C \quad \because B\bar{B}=0 \\
 &= A\bar{B}CC + \bar{A}\bar{B}C \\
 &= A\bar{B}C + \bar{A}\bar{B}C \quad \because CC=C \\
 &= \bar{B}C(A + \bar{A}) \\
 &= \bar{B}C(1) \\
 &= \bar{B}C
 \end{aligned}$$

Simplify the following Boolean expression.

$$\begin{aligned}
 & \overline{A}BC + \overline{A}\overline{B}\overline{C} + \overline{A}\overline{B}C + \overline{A}B\overline{C} + A\overline{B}C + A\overline{B}\overline{C} \\
 &= \overline{A}BC + \overline{A}\overline{B}\overline{C} + \overline{A}\overline{B}C + \overline{A}B\overline{C} + A\overline{B}C + A\overline{B}\overline{C} \\
 &= \overline{A}BC + A\overline{B}C + \overline{A}\overline{B}\overline{C} + \overline{A}\overline{B}C + \overline{A}B\overline{C} + A\overline{B}\overline{C} \\
 &= BC(A + \overline{A}) + \overline{A}\overline{B}(\overline{C} + C) + \overline{A}B\overline{C} + A\overline{B}\overline{C} \\
 &= BC(1) + \overline{A}\overline{B}(1) + \overline{A}B\overline{C} + A\overline{B}\overline{C} \\
 &= BC + \overline{A}\overline{B} + \overline{A}B\overline{C} + A\overline{B}\overline{C} \\
 &= BC + \overline{B}(A + \overline{A}\overline{C}) + 1 \quad \text{Wrong} \\
 &= BC + \dots
 \end{aligned}$$

Simplify the following Boolean expression.

$$\begin{aligned}
 & \overline{A}BC + \overline{A}\overline{B}\overline{C} + \overline{A}\overline{B}C + \overline{A}B\overline{C} + A\overline{B}C + A\overline{B}\overline{C} \\
 &= \overline{A}BC + A\overline{B}C + \overline{A}\overline{B}\overline{C} + \overline{A}\overline{B}C + \overline{A}B\overline{C} + A\overline{B}\overline{C} \\
 &= BC(A + \overline{A}) + \overline{A}\overline{B}(\overline{C} + C) + \overline{A}B\overline{C} + A\overline{B}\overline{C} \\
 &= BC(1) + \overline{A}\overline{B}(1) + \overline{A}B\overline{C} + A\overline{B}\overline{C} \\
 &= BC + \overline{A}\overline{B} + \overline{A}B\overline{C} + A\overline{B}\overline{C} \\
 &= BC + \overline{B}(A + \overline{A}\overline{C}) \\
 &\therefore A + \overline{A}\overline{C} = A \\
 &= BC + \overline{A}\overline{B} + \overline{B}\overline{C}
 \end{aligned}$$

Example 4-12

Simplify the following Boolean expression.

$$\overline{AB} + AC + \overline{A}BC$$

Apply De Morgan's theorem

$$(\overline{A}\overline{B})(\overline{A}C) + \overline{A}BC$$

Apply De Morgan's theorem each in parentheses:-

$$\begin{aligned}
 & (\overline{A} + \overline{B})(\overline{A} + \overline{C}) + \overline{A}BC \\
 &= \overline{A}\overline{A} + \overline{A}\overline{C} + \overline{B}\overline{A} + \overline{B}\overline{C} + \overline{A}BC \\
 &= \overline{A} + \overline{A}\overline{C} + \overline{A}\overline{B} + \overline{B}\overline{C} + \overline{A}BC \\
 &= \overline{A} + \overline{A}\overline{C} + \overline{A}\overline{B}(1) + \overline{B}\overline{C} \\
 &= \overline{A} + \overline{A}\overline{C} + \overline{A}\overline{B} + \overline{B}\overline{C} \\
 &= \overline{A}(1 + \overline{C}) + \overline{A}\overline{B} + \overline{B}\overline{C} \\
 &= \overline{A}\overline{A} + \overline{A}\overline{B} + \overline{B}\overline{C} \\
 &= \overline{A}(1 + \overline{B}) + \overline{B}\overline{C} \\
 &= \overline{A} \cdot \overline{A} + \overline{B}\overline{C} \\
 &= \overline{A} + \overline{B}\overline{C}
 \end{aligned}$$

Related problem:
Simplify the following Boolean expression

$$\begin{aligned} & \overline{A}B + \overline{A}C + \overline{A}\overline{B}\overline{C} \\ &= (\overline{A} + \overline{B}) + (\overline{A} + \overline{C}) + \overline{A}\overline{B}\overline{C} \\ &= \overline{A} + \overline{B} + \overline{A} + \overline{C} + \overline{A}\overline{B}\overline{C} \\ &= \overline{A}B + \overline{A}C + (\overline{A} + \overline{B} + \overline{C}) \\ &= \overline{A} \end{aligned}$$

$$\begin{aligned} & \overline{A}B + \overline{A}C + \overline{A}\overline{B}\overline{C} \\ &= \overline{A} + \overline{B} + \overline{A} + \overline{C} + \overline{A}\overline{B}\overline{C} \\ &= \overline{A} + \overline{B} + \overline{C} + \overline{A}\overline{B}\overline{C} \\ &= \overline{A}(1 + \overline{B}\overline{C}) + \overline{C} + \overline{B} \\ &= \overline{A} + \overline{B} + \overline{C} \end{aligned}$$

Implicant:

An implicant is the product of some of the variables (in complemented or uncomplemented form) for which the function is equal to one.

Simplify the following boolean expression.

$$\begin{aligned} \text{(i)} \quad & \overline{A}B + AB + ABC + \overline{A}\overline{B}C \\ &= \overline{A}B + AB(1+C) + \overline{A}\overline{B}C \\ &= \overline{A}B + AB(1) + \overline{A}\overline{B}C \\ &= \overline{A}B + AB + \overline{A}\overline{B}C \\ &= A(\overline{B} + B) + \overline{A}\overline{B}C \\ &= A(1) + \overline{A}\overline{B}C \\ &= A + \overline{A}\overline{B}C \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad & \overline{A}BC + A\overline{C} + \overline{A}\overline{B}\overline{C} + ABC \\ &= BC(\overline{A} + A) + A\overline{C} + \overline{A}\overline{B}\overline{C} \\ &= BC(1) + A\overline{C} + \overline{A}\overline{B}\overline{C} \\ &= BC + A\overline{C} + \overline{A}\overline{B}\overline{C} \\ &= BC + \overline{C}(A + \overline{A}\overline{B}) \\ &= BC + \overline{C}(A + \overline{A})(A + \overline{B}) \\ &= BC + \overline{C}(1)(A + \overline{B}) \\ &= BC + A\overline{C} + \overline{B}\overline{C} \end{aligned}$$

Prove that $P + \overline{P}Q = P + Q$

$$\begin{aligned} & P + \overline{P}Q = P + Q \\ & (P + \overline{P})(P + Q) = P + Q \quad \therefore A + AB = (A + \overline{A})A + A \\ & (1)(P + Q) = P + Q \\ & P + Q = P + Q \end{aligned}$$

(iii)

$$XYZ + XY\bar{Z} + X\bar{Y} + Y\bar{Z}$$

$$= XY(Z + \bar{Z}) + X\bar{Y} + Y\bar{Z}$$

$$= XY(1) + X\bar{Y} + Y\bar{Z}$$

$$= XY + X\bar{Y} + Y\bar{Z}$$

$$= X(Y + \bar{Y}) + Y\bar{Z}$$

$$= X(1) + Y\bar{Z}$$

$$= X + Y\bar{Z}$$

(iv) $\bar{P}\bar{Q}R + \bar{P}R + P\bar{Q}$

$$= \bar{P}\bar{Q}R + \bar{P}R + P\bar{Q}$$

$$= \bar{P}\bar{Q}(R + 1) + \bar{P}R$$

$$= \bar{P}\bar{Q} + \bar{P}R$$

(v) $A + AB + A\bar{B}C$

$$= A + AB + A\bar{B}C$$

$$= A(1 + B) + A\bar{B}C$$

$$= A + A\bar{B}C$$

$$= A(1 + \bar{B}C)$$

$$= A(1)$$

$$= A$$

(vi) $(\bar{A} + B)C + ABC$

$$= \bar{A}C + BC + ABC$$

$$= \bar{A}C + BC(1 + A)$$

$$= \bar{A}C + BC$$

$$= C(\bar{A} + B)$$